

Absolutely. For **Experiment 8 — SOM Clustering**, here is a proper **lab-record write-up** with Aim, Objectives, Theory, Working Principle, Algorithm, Program, Expected Output, Result, Conclusion, and Viva questions. I'll keep the theory formal enough for submission without making it unnecessarily bloated.

EXPERIMENT 8

Implementation of Cluster Data with a Self-Organizing Map using MATLAB

1. Aim

To implement clustering of multidimensional data using a **Self-Organizing Map (SOM)** in MATLAB and visualize the clusters formed by the network.

3. Software Requirements

- MATLAB
 - Deep Learning Toolbox / Neural Network Toolbox
 - Computer system with Windows/Linux/macOS
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4. Theory

A **Self-Organizing Map (SOM)**, also known as a **Kohonen Map**, is an **unsupervised artificial neural network** used for clustering, dimensionality reduction, visualization, pattern recognition, and feature extraction.

Unlike supervised learning algorithms, a SOM does not require predefined target labels. It learns the underlying structure of the input data based on the similarity between data samples.

A SOM consists of a set of neurons arranged in a usually **two-dimensional grid**. Each neuron has a weight vector of the same dimension as the input data. During training, the network compares an input vector with the weight vectors of all neurons and identifies the neuron whose weight is most similar to the input. This neuron is called the **Best Matching Unit (BMU)**.

The BMU and its neighboring neurons are then updated so that their weights become closer to the input vector. This process is repeated for many training iterations. As training progresses, similar input samples tend to activate nearby neurons on the SOM grid.

Therefore, a SOM performs clustering while also preserving the **topological relationships** present in the original data. In other words, data points that are similar in the original feature space are generally mapped to nearby neurons on the two-dimensional grid.

5. Working Principle

The working of a Self-Organizing Map can be explained as follows:

Step 1: Initialization

The weight vectors of all neurons in the SOM are initialized.

Step 2: Select an Input

An input vector is selected from the training dataset.

Step 3: Calculate Distance

The distance between the input vector and every neuron's weight vector is calculated. **Euclidean distance** is commonly used.

$$d_i = \sqrt{\sum_j (x_j - w_{ij})^2}$$

where:

- x_j = input vector component
- w_{ij} = weight of neuron i
- d_i = distance between input and neuron i

Step 4: Find the BMU

The neuron having the minimum distance from the input vector is selected as the **Best Matching Unit (BMU)**.

$$BMU = \arg\min_i (d_i)$$

Step 5: Update Weights

The weight of the BMU and its neighboring neurons is updated toward the input vector.

$$w_i(t+1) = w_i(t) + \alpha(t) h_{ci}(t) [x(t) - w_i(t)]$$

where:

- $\alpha(t)$ = learning rate
- $h_{ci}(t)$ = neighborhood function
- $x(t)$ = current input
- $w_i(t)$ = current weight

Step 6: Repeat Training

The above process is repeated for multiple input samples and training iterations.

After training, similar samples are mapped to neighboring neurons, producing a meaningful representation of the dataset on the SOM grid.

6. Dataset Used

In this experiment, three artificial clusters are generated using MATLAB.

- **Cluster 1:** Centered approximately around (2, 2)
- **Cluster 2:** Centered approximately around (7, 3)
- **Cluster 3:** Centered approximately around (4, 8)

Each cluster contains **100 data points**, giving a total of **300 samples**.

The data is generated using random values with a fixed random seed so that the experiment can be reproduced.

7. Algorithm

1. Start the MATLAB program.
2. Generate or load the dataset.
3. Normalize the data if required.
4. Define the SOM grid size.
5. Create the SOM network.
6. Train the SOM using the input dataset.
7. Find the Best Matching Unit for every input sample.
8. Assign each sample to its corresponding SOM neuron.
9. Visualize the SOM topology and sample distribution.
10. Analyze the cluster distribution.
11. Stop.

11. Conclusion

A **Self-Organizing Map** is an effective unsupervised neural-network technique for **clustering and visualization**. In this experiment, a 5×5 SOM was trained using artificially generated clustered data. The network identified different regions of the input space and represented them using neighboring neurons on a two-dimensional grid.

Thus, SOM provides both **clustering capability and topology-preserving visualization** of input data.